

JASON ALEJANDRO MENDOZA MONTROYA

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EDUCATION

COLUMBIA UNIVERSITY

New York, NY

Bachelor of Science in Computer Science

May 2025

Relevant Coursework: Advanced Software Engineering, Design for Generative AI, Fundamentals of Computer Systems, Linear Algebra, UI Design, Intro to Databases, Data Structures and Algorithms, Advanced Programming in C, Discrete Mathematics, Computer Science Theory, Calculus III.

EXPERIENCE

HEALTHFIRST

New York, NY

Software Engineer

Sept 2025 – Present

- Architected end-to-end backend infrastructure for enterprise Data Decision Management platform, leveraging serverless AWS services (Lambda, API Gateway, RDS, CloudWatch) and relational data models.
- Reduced workflow runtime from 35 minutes to under 5 minutes, driving an 85%+ processing efficiency improvement across enterprise operations.
- Presented platform Minimum Viable Product (MVP) and system architectural benchmarks directly to Chief Technology Officer and senior leadership.

COLUMBIA BUILD LAB - PARTHION BATTERY SOLUTIONS

New York, NY

Lead Software Engineer Intern

Jan 2025 – May 2025

- Spearheaded full-stack development of startup MVP, delivering core front-end UI and back-end database capabilities for scalable web-based marketplace.
- Engineered cross-functional features in Agile environment to ensure seamless deployment and reliable system modularity.

NATIONAL GRID

New York, NY

Software Engineer Intern

June 2024 – Aug 2024

- Increased ticket resolution velocity by 400% (handling up to 5 full-stack tickets weekly vs. 1 previously), raising team productivity by 30%.
- Developed full-stack features using TypeScript and Node.js; refactored codebases to resolve cognitive complexity flags via SonarQube.
- Enforced rigorous software quality standards utilizing Git version control, unit testing frameworks, code reviews, and Agile Jira workflows.

PROJECTS

CHEATCODE – AI LEETCODE PROBLEM SOLVER

Flask, OpenAI API, JavaScript, Bootstrap

- Engineered web application integrating OpenAI GPT API to automate problem-solving across 200+ distinct LeetCode algorithm categories.
- Achieved sub-30-second execution times with 98% programmatic accuracy while cutting technical interview prep time for users by 50%.
- Implemented feedback instrumentation to log accuracy tracking and user ratings for continuous prompt and output refinement.

VOLUNTEER & DONATION MANAGEMENT PLATFORM

Java, Spring Boot, Google Cloud SQL, GCP

- Built cloud-native Spring Boot application and Google Cloud SQL infrastructure to streamline nonprofit volunteer management and donation tracking.
- Designed 30+ RESTful API endpoints for secure user authentication, real-time scheduling, and automated inventory tracking.
- Achieved 95%+ unit and integration test coverage with JaCoCo and Maven, automated via GitHub Actions CI/CD to reduce production bugs by 30%.

SKILLS & INTERESTS

Languages & Frameworks: Java, C, Python, TypeScript, JavaScript, HTML/CSS, React, Node.js, Express, Flask, Spring Boot, Tailwind CSS, Bootstrap

Cloud & Developer Tools: AWS (Lambda, API Gateway, RDS, CloudWatch, VPC, IAM), GCP, Git, GitHub Actions, SonarQube, Postman, Jira, Maven, JaCoCo

Databases & AI Integration: PostgreSQL, MySQL, Google Cloud SQL, OpenAI API, ElevenLabs, Whisper, Google Calendar API, Jupyter